Example

If the retention factor for a component is k=5, what fraction of time does this component spend in the mobile phase?

$$k=5=(t_r-t_m)/t_m = t_s/t_m$$

Fraction spent in mobile phase = $t_m/(t_m+t_s)$ =?

$$t_m/(t_m+t_s) = t_m/(t_m+5t_m) = t_m/6t_m = 1/6$$

Chromatographic Resolution, R

A certain gas chromatogram showed an air peak, which was a marker for t_m, at 30.0 s and two poorly resolved peaks at 70.0 s and 90.0 s, respectively. All times were measured from injection. The two poorly resolved peaks had base widths of 13.0 s and 15.0 s, respectively. Calculate R for the two peaks and estimate the % resolution using the normal distribution table.

0z0	y	Area	0z0	y	Area
1.4	0.149 7	0.419 2	2.8	0.007 9	0.497 4
1.5	0.129 5	0.433 2	2.9	0.006 0	0.498 1
1.6	0.110 9	0.445 2	3.0	0.004 4	0.498 650
1.7	0.094 1	0.455 4	3.1	0.003 3	0.499 032
1.8	0.079 0	0.464 1	3.2	0.002 4	0.499 313
1.9	0.065 6	0.471 3	3.3	0.001 7	0.499 517

$R = \Delta t / w_{av} = 2(90.0-70.0)/(13.0+15.0) = 1.428..$

$z = 2R = 2.857.. \sim 2.9$

Area resolved = 0.5+0.4981 = 0.9981 i.e. 99.81% resolved